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Singular Value Decomposition

Want : $A = U \Sigma V^T$

$$\boxed{A = \begin{pmatrix} 5 & 5 \\ -1 & 7 \end{pmatrix}}$$

We know : * $A^TA = (U\Sigma V^T)^T (U \Sigma V)$

$= V \Sigma^T U^T U \Sigma V$

$= V \Sigma^T \Sigma V$

* $AV = U\Sigma$

Step 1. Find A^TA

$$A^TA = \begin{pmatrix} 5 & -1 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 5 & 5 \\ -1 & 7 \end{pmatrix} = \begin{pmatrix} 26 & 18 \\ 18 & 74 \end{pmatrix}$$

Step 2. Find the eigenvalues of A^TA

$AX = \lambda X$

$\rightarrow AX - \lambda X = 0$

$\rightarrow (A - \lambda) X = 0$

$\therefore |A - \lambda| = 0$

$$\begin{vmatrix} 26-\lambda & 18 \\ 18 & 74-\lambda \end{vmatrix}$$

$\rightarrow (26-\lambda)(74-\lambda) - 18^2 = 0$

$\rightarrow \lambda^2 + 1924 - 100\lambda - 324 = 0$

$\rightarrow \lambda^2 - 100\lambda + 1600 = 0$

$\rightarrow (\lambda - 20)(\lambda - 80) = 0$

$\therefore \lambda_1 = 20$ & $\lambda_2 = 80$

Step 3. Find the eigen vectors for the eigen values

For 20 : $A - 20I = \begin{pmatrix} 6 & 18 \\ 18 & 54 \end{pmatrix}$

Now, $(A - \lambda I) X = 0$ $\therefore \begin{pmatrix} 6 & 18 \\ 18 & 54 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\therefore x_1 = -3 \quad x_2 = 1$

$$v_1 = \begin{pmatrix} -3 \\ 1 \end{pmatrix} = \begin{pmatrix} -3/\sqrt{10} \\ 1/\sqrt{10} \end{pmatrix}$$

Normalized $\left(\sqrt{(-3)^2 + (1)^2}\right)$

For 80 :

$$A - 80I = \begin{pmatrix} -64 & 18 \\ 18 & -6 \end{pmatrix} \qquad v_2 = \begin{pmatrix} 1/\sqrt{10} \\ 3/\sqrt{10} \end{pmatrix}$$

Step 4. Get V and Σ

$$V = \begin{pmatrix} | & | \\ v_1 & v_2 \\ | & | \end{pmatrix} = \begin{pmatrix} \frac{-3}{\sqrt{10}} & \frac{1}{\sqrt{10}} \\ \frac{1}{\sqrt{10}} & \frac{3}{\sqrt{10}} \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{pmatrix} = \begin{pmatrix} \sqrt{20} & 0 \\ 0 & \sqrt{80} \end{pmatrix} = \begin{pmatrix} 2\sqrt{5} & 0 \\ 0 & 4\sqrt{5} \end{pmatrix}$$

Step 5. Use $AV = U\Sigma$ to get U

$$AV = \begin{pmatrix} 5 & 5 \\ -1 & 7 \end{pmatrix} \begin{pmatrix} -3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & 3/\sqrt{10} \end{pmatrix} = \begin{pmatrix} -\sqrt{10} & 2\sqrt{10} \\ \sqrt{10} & 2\sqrt{10} \end{pmatrix}$$

Now, we know $U\Sigma = \begin{pmatrix} -\sqrt{10} & 2\sqrt{10} \\ \sqrt{10} & 2\sqrt{10} \end{pmatrix}$

therefore we divide to cols by $\sqrt{\lambda_1}$ and $\sqrt{\lambda_2}$

$$U = \begin{pmatrix} \frac{-\sqrt{10}}{2\sqrt{5}} & \frac{2\sqrt{10}}{4\sqrt{5}} \\ \frac{\sqrt{10}}{2\sqrt{5}} & \frac{2\sqrt{10}}{4\sqrt{5}} \end{pmatrix} = \begin{pmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

Principal Component Analysis

$X^T =$

x_1	4	8	13	7
x_2	11	4	5	14

Step 1. Get the means.

$$\bar{x}_1 = 32/4 = 8$$
$$\bar{x}_2 = 34/4 = 8.5$$

Step 2. Get the data points to the origin

$\hat{x}_1 \rightarrow x_1 - \bar{x}_1 \qquad \hat{x}_2 \rightarrow x_2 - \bar{x}_2$

$\hat{X}^T =$

$\hat{x}_1$	-4	0	5	-1
$\hat{x}_2$	2.5	-4.5	-3.5	5.5

Step 3. Get the covariance matrix

$\because$ mean $= 0 \qquad Cov = \frac{1}{N-1} \hat{X}^T X$

$$\frac{1}{3} \hat{X}^T X = \begin{pmatrix} -4 & 0 & 5 & -1 \\ \frac{5}{2} & -\frac{9}{2} & -\frac{7}{2} & \frac{11}{2} \end{pmatrix} \begin{pmatrix} -4 & 5/2 \\ 0 & -9/2 \\ 5 & -7/2 \\ -1 & 11/2 \end{pmatrix} \times \frac{1}{3}$$

$$= \begin{pmatrix} 42 & -33 \\ -33 & 69 \end{pmatrix} \times \frac{1}{3} = \begin{pmatrix} 14 & -11 \\ -11 & 23 \end{pmatrix}$$

Step 4. Get the Eigen Values

$$|A - \lambda I| = 0 \rightarrow \begin{pmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{pmatrix} = 0$$

$$\rightarrow \lambda^2 - 37\lambda + 20 = 0$$

$$\therefore \lambda_1 = 30.3849 \qquad \lambda_2 = 6.6151$$

Step 4. Get the Eigen vectors

$$\begin{pmatrix} 14-\lambda & -11 \\ -11 & 23-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$\rightarrow (14-\lambda)x_1 - 11x_2 = 0$ and $-11x_1 + (23-\lambda)x_2 = 0$

For $\lambda_1 = 30.3849$ $\quad v_1 = \begin{pmatrix} 11 \\ -16.3849 \end{pmatrix}$

Step 5. Get the unit Eigen vector

$$|V_1| = \sqrt{(11)^2 + (-16.3849)^2} = 19.7348$$

$$\therefore e_1 = \frac{v_1}{19.7348} = \begin{pmatrix} 0.5574 \\ -0.8303 \end{pmatrix}$$

Step 6. The first principal Component

$$FPC = e_1^T \begin{pmatrix} \hat{x}_{1k} \\ \hat{x}_{2k} \end{pmatrix}$$

For $k=1$: $x_{1k} = -4$ $\quad x_{2k} = 5/2$

$$FPC = \begin{pmatrix} 0.55 & -0.83 \end{pmatrix} \begin{pmatrix} -4 \\ 5/2 \end{pmatrix} = -4.3053$$

For $k=2$: $FPC = 3.7361$

For $k=3$: $FPC = 5.6928$

For $k=4$: $FPC = -5.1238$